

SIBOLGA PINANGSORI, Indonesia

WMO: 960730

Lat: 1.556N

Lon: 98.889E

Elev: 43

StdP: 14.67

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	70.4	71.3	68.1	103.4	78.0	69.3	107.9	77.5	13.0	83.7	11.1	84.5	0.1	N/A	0.259

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.4	91.7	78.6	90.8	78.5	90.0	78.4	81.7	88.0	81.0	87.3	80.4	86.8	5.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.9	155.5	85.7	79.2	151.7	85.1	78.7	149.0	84.6	45.5	88.4	44.8	87.6	44.1	87.1	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.7	9.8	8.7	DB	66.0	94.3	3.5	1.2	63.4	95.2	61.4	95.9	59.4	96.5	56.8	97.4
			WB	65.0	83.9	3.0	1.5	62.8	85.0	61.1	85.8	59.4	86.7	57.2	87.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	80.6	81.0	81.3	81.5	81.7	81.0	80.1	80.0	80.1	79.9	79.8	80.2	(6)
	DBStd	1.71	1.56	1.61	1.60	1.57	1.56	1.59	1.69	1.62	1.62	1.51	1.53	1.53	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11171	948	869	971	945	982	930	934	931	903	928	894	936	(10)
	CDD65	5696	483	449	506	495	517	480	469	466	453	463	444	471	(11)
	CDH74	50127	4265	4101	4691	4629	5075	4481	4112	3892	3800	3746	3427	3908	(12)
	CDH80	17603	1534	1570	1769	1656	1926	1696	1497	1339	1238	1144	985	1249	(13)
(14) Wind	WSAvg	1.3	1.2	1.3	1.4	1.2	1.3	1.4	1.4	1.4	1.3	1.3	1.2	1.1	(14)
(15) Precipitation	PrecAvg	160.5	10.9	11.0	15.6	15.1	11.1	9.1	12.1	10.4	14.7	17.3	17.6	16.1	(15)
	PrecMax	195.7	20.0	19.2	26.7	22.5	27.8	19.0	23.6	21.9	25.6	33.6	31.1	29.4	(16)
	PrecMin	128.1	1.9	2.2	3.2	8.0	2.9	2.7	4.1	2.4	7.6	4.8	8.7	7.4	(17)
	PrecStd	18.1	4.3	4.2	5.6	3.6	4.8	3.7	5.2	4.4	4.1	5.7	5.7	5.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.1	92.9	92.7	91.8	91.9	91.8	91.1	90.7	90.7	90.0	89.6	90.6	(19)
		MCWB	78.0	77.9	79.5	80.1	79.2	78.3	77.3	77.4	77.6	77.4	78.2	78.0	(20)
	2%	DB	90.2	90.9	91.0	90.4	90.7	90.6	89.8	89.4	89.3	88.7	88.1	89.1	(21)
		MCWB	78.1	78.3	79.1	79.6	78.9	78.2	77.5	77.5	77.7	77.6	78.1	78.2	(22)
	5%	DB	88.9	89.7	89.7	89.2	89.7	89.4	88.7	88.2	88.1	87.3	86.9	87.8	(23)
		MCWB	78.0	78.4	78.9	79.3	78.8	78.2	77.6	77.7	77.7	77.7	78.1	78.2	(24)
	10%	DB	87.4	88.3	88.2	87.9	88.5	88.1	87.3	86.8	86.5	85.9	85.4	86.4	(25)
		MCWB	77.8	78.3	78.7	79.1	78.9	78.2	77.7	77.7	77.7	77.7	77.9	78.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	82.2	82.2	82.3	81.9	81.6	81.0	80.7	80.6	80.6	81.0	81.0	(27)
		MCDB	87.8	88.3	89.1	88.7	88.2	88.1	87.0	86.6	86.5	86.3	86.4	86.8	(28)
	2%	WB	80.1	80.9	81.0	81.2	81.0	80.4	79.9	79.5	79.6	79.5	79.8	79.9	(29)
		MCDB	86.6	87.4	87.9	87.6	87.3	86.9	86.2	85.9	85.6	85.2	85.3	85.8	(30)
	5%	WB	79.2	79.9	80.1	80.5	80.2	79.7	79.1	78.9	78.9	78.8	79.0	79.1	(31)
		MCDB	85.8	86.6	86.9	86.9	86.6	86.2	85.4	85.4	85.0	84.5	84.5	85.1	(32)
	10%	WB	78.5	79.0	79.3	79.7	79.6	79.0	78.3	78.2	78.2	78.1	78.3	78.5	(33)
		MCDB	84.9	85.8	86.0	85.9	85.9	85.6	84.6	84.5	84.1	83.6	83.5	84.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	15.4	14.9	13.8	14.4	14.8	14.8	14.1	13.4	12.8	12.2	13.2	(35)
		MCDBR	16.8	17.3	16.8	15.4	16.0	16.7	16.9	16.2	15.7	15.0	14.2	15.4	(36)
	5% WB	MCWBR	6.8	7.0	6.8	6.3	6.3	6.7	6.9	6.6	6.1	6.0	6.0	6.3	(37)
		MCWBR	15.0	15.7	15.5	14.5	14.8	15.2	15.1	14.9	14.2	13.2	13.0	13.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.458	0.535	0.506	0.446	0.434	0.440	0.432	0.462	0.467	0.501	0.442	0.423	(40)
		taud	2.332	2.087	2.196	2.378	2.424	2.412	2.430	2.328	2.311	2.200	2.387	2.442	(41)
	Ebn at Noon		276	257	264	274	270	265	268	266	270	263	279	285	(42)
		Edh at Noon	42	54	49	40	36	36	36	41	43	48	39	37	(43)
(44) All-Sky Solar Radiation	RadAvg		1583	1676	1709	1650	1655	1601	1585	1567	1527	1509	1477	1501	(44)
	RadStd		69	94	94	68	76	62	83	79	117	186	62	66	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page